

Load CSV file and print dataset statistics using R

Author: Marcos de Carvalho

Date: 2026-04-25

Loads a CSV file using R's `read.csv` and prints rows, columns, total cells, file size, and object memory usage.

Language

rscript

Category

data-processing

Command

```
Rscript -e "args<-commandArgs(trailingOnly=TRUE);if(length(
  args)<1)stop('Need filename');df<-read.csv(args[1]);cat('
  Rows:',nrow(df),'Cols:',ncol(df),'Cells:',nrow(df)*ncol(
  df),'File size:',file.size(args[1]),'Object size:',format
  (object.size(df),units='auto'),'\\n')"
```

Explanation

The command uses Rscript to execute R code that reads a CSV file specified as argument, computes basic statistics (rows, columns, cells), gets file size via `file.size()`, and reports object memory usage via `object.size()`.

Tags

rscript, csv, statistics, data-analysis, memory

Dependencies

r-base-core

Arguments

1. **CSV_FILE** (Required): CSV file to analyze

Examples

1. `Rscript -e "args<-commandArgs(trailingOnly=TRUE);if(length(args)<1)stop('Need filename');df<-read.csv(args[1]);cat('Rows:',nrow(df),'Cols:',ncol(df),'Cells:',nrow(df) size:',file.size(args[1]),'Object size:',format(object.size(df),units='auto'),'\\{\\}n')"`
data.csv - Analyze data.csv
2. `Rscript -e "args<-commandArgs(trailingOnly=TRUE);if(length(args)<1)stop('Need filename');df<-read.csv(args[1]);cat('Rows:',nrow(df),'Cols:',ncol(df),'Cells:',nrow(df) size:',file.size(args[1]),'Object size:',format(object.size(df),units='auto'),'\\{\\}n')"`
large_dataset.csv 2>/dev/null - Analyze suppressing warnings

Output

```
Rows: 100 Cols: 10 Cells: 1000 File size: 12345 Object size:
78.2 Kb
```

Notes

- Uses read.csv with default settings (header=TRUE, stringsAsFactors=FALSE in R_i=4.0)
- File size reported in bytes
- Object size includes R overhead
- For large files, consider adding options(stringsAsFactors=FALSE)

Warnings

- May load entire file into memory
- Very large CSV files could exhaust memory

See Also

- tsv-stats

Status

draft

Safety

safe

Shell

posix

Platforms

linux, gnu-linux, freebsd, openbsd, netbsd

Created At

2026-04-25

Updated At

2026-04-25